

KUSHAGRA SINGHAL

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SUMMARY

Backend Engineer with ownership of 5 SaaS platforms. Specializing in FastAPI microservices, PostgreSQL optimization, and distributed system design.

WORK EXPERIENCE

Founding Engineer – Datavedam, Bangalore

Dec 2025 – Present

- Architected 5 SaaS platforms on FastAPI and PostgreSQL, serving 500+ users.
- Designed multi-service backend with schema isolation, cutting releases by 40%.
- Developed 150+ REST APIs with Pydantic validation, RBAC, and exception handling.
- Normalized PostgreSQL schemas with pgvector indexing, lowering latency by 35%.
- Automated CI/CD pipelines (GitHub Actions, pytest, Docker), reaching 80%+ coverage.

Data Analyst Intern – Nineleaps, Bangalore

Mar 2025 – Aug 2025

- Optimized SQL queries on 1M+ records, boosting performance by 35%.
- Reduced BI dashboard refresh time by 40% by eliminating redundant joins.
- Configured 3 CI/CD workflows with pytest validation, decreasing defects by 25%.

PROJECTS

BroAI – AI Agent Memory System

- Engineered unified PostgreSQL store replacing 3 databases, cutting costs by 60%.
- Created hybrid search API (pgvector + BM25 + RRF) with 10ms query latency.
- Implemented PII masking and Transactional Outbox for ACID-compliant writes.
- **Tech:** FastAPI, PostgreSQL, pgvector, SQLAlchemy, Redis, Docker

ShikshaVedam – AI-First School ERP

- Built RAG pipeline with pgvector, powering 4 AI agent workflows.
- Structured multi-tenant schema with Keycloak RBAC serving 500+ users.
- Decreased administrative workload by 40%, improving report speed by 3x.
- **Tech:** FastAPI, PostgreSQL, pgvector, Keycloak, Docker

PracharVedam – Ad Analytics Platform

- Refactored monolith into 5 microservices with isolated PostgreSQL schemas.
- Deployed event-driven analytics with TimescaleDB, reducing response time by 28%.
- Integrated Google Ads and Meta Ads APIs for cross-platform campaign reporting.
- **Tech:** FastAPI, PostgreSQL, TimescaleDB, Redis, Docker

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend:** FastAPI, Pydantic, REST APIs, WebSockets, Microservices Architecture
- **Databases:** PostgreSQL, pgvector, Redis, SQLAlchemy, Query Optimization, Indexing
- **Architecture:** Distributed Systems, Event-Driven Architecture, Transactional Outbox Pattern, RBAC, JWT
- **DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD Pipelines
- **Testing:** pytest, Robot Framework, Unit and Integration Testing
- **Practices:** SOLID Principles, OOP, Design Patterns, Agile and Scrum
- **Leadership:** Cross-functional collaboration, team mentoring, stakeholder communication, sprint planning

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2021 – 2025

Christ University, Bangalore

CERTIFICATIONS

- **Google Data Analytics Professional Certificate** – Coursera
- **AI Agents with LangChain and LangGraph** – IBM